

EMBRYOLOGY

Embryology = the study of **prenatal development**, from **fertilization** → **formation of the embryo** → **fetal growth** → **birth**.

It explains how a single cell ☺ develops into a complex human being through coordinated **cell division, migration, differentiation, growth, and organ formation**.

TWO MAJOR PERIODS OF PRENATAL DEVELOPMENT

1 **Embryonic period**

Week 1 → end of Week 8

This is the period of **embryogenesis and organogenesis**.

During this period:

☺ **Zygote**



⊗ **Morula**



○ **Blastocyst**



 **Embryo**



♥ ☹ **Development of major organ systems**

★ **Most important point:**

Embryonic period = organ formation

This is the **most sensitive period for teratogens** because organs are being formed.

2 **Fetal period** ☹

Week 9 → birth

The basic organ systems have already been established.

The major processes are:

 **Growth**

 **Differentiation and maturation**

 **Increase in body weight**

Functional maturation of organs

★ **Most important point:**

Fetal period = growth + maturation



Weeks 1–8 = **MAKE** the organs

Week 9–birth = **GROW** the organs

TERATOLOGY

Teratology = the study of the **causes, mechanisms, and effects of congenital abnormalities (birth defects)**.

Teratogens

Teratogens are environmental agents that can cause abnormal development.

Examples include:

- Certain drugs
- Alcohol
- Some infections
- Radiation
- Certain chemicals
- Nutritional deficiencies

The effect of a teratogen depends on:

Timing + Dose + Duration + Genetic susceptibility

Timing is especially important.

During **organogenesis**, exposure can produce major structural abnormalities.

WHY IS EMBRYOLOGY IMPORTANT?

Clinical medicine

Embryology helps explain conditions encountered in:

- Obstetrics**
- Pediatrics**
- Perinatal medicine**
- Clinical anatomy**
- Genetics**

Understanding congenital abnormalities

Knowing normal development helps us understand **how congenital malformations occur**.

For example:

- ♥ Abnormal cardiac development → congenital heart defects
- 🧠 Abnormal neural tube development → neural tube defects

3 Prenatal diagnosis 📖

Embryological knowledge helps doctors understand:

- ✓ Ultrasound findings
- ✓ Prenatal screening
- ✓ Congenital abnormalities
- ✓ Fetal development

4 Infertility and reproductive medicine 🧠

Understanding:

- 🧬 Gametogenesis
- ➡ Fertilization
- ➡ Implantation
- ➡ Early embryonic development

is important in **fertility treatment and assisted reproductive technologies**.

📖 MODERN EMBRYOLOGY

Embryology has evolved from simply observing developing embryos 🔍 to studying the **molecular mechanisms controlling development**.

⚠ **Embryonic period is the most critical period for major structural congenital abnormalities because organogenesis is occurring.**

◇ One small terminology point: **embryogenesis** is broader than organogenesis; organogenesis specifically refers to the formation of organs, mainly during the embryonic period.

[24/09/2026 6:55 AM] Imaanaa Harmee:

🧬 GAMETOGENESIS

Gametogenesis = the process by which **primordial germ cells (PGCs)** develop into mature **male gametes (spermatozoa)** and **female gametes (oocytes)**.

⚡ **Main purpose:** Reduce chromosome number from **diploid (46, 2n) → haploid (23, n)** through **meiosis**, so that fertilization restores the normal **46 chromosomes**.

🔑 Four major phases

1, **Migration** – Primordial germ cells migrate to the developing gonads.

- 2, **Multiplication** – Germ cells increase in number by **mitosis**.
- 3, **Reduction** – Chromosome number is reduced by **meiosis**.
- 4, **Maturation** – Cells undergo structural and functional specialization to form sperm or oocytes.

SPERMATOGENESIS

Spermatogenesis = formation of mature spermatozoa from spermatogonia.

 **Site:** Seminiferous tubules of the testes

 **Begins:** At **puberty**

 **Continuous:** Continues throughout adult life, although it may decline with age.

 Sequence

Spermatogonium → **Primary spermatocyte** → **Secondary spermatocyte** → **Spermatid**
→ **Spermatozoon**

1 **Spermatogonia**

Primordial germ cells eventually give rise to **spermatogonia**.

There are two important types:

 **Type A spermatogonia** → stem cells; maintain the germ-cell population.

Type B spermatogonia → differentiate and enter meiosis.

2 **Primary spermatocyte**

Type B spermatogonium replicates its DNA and becomes a:

 **Primary spermatocyte**

It is **diploid** ($2n = 46$) but has duplicated DNA.

Then it enters **Meiosis I**.

3 **Secondary spermatocytes**

1 primary spermatocyte

 Meiosis I

2 secondary spermatocytes

Each secondary spermatocyte is:

 **Haploid = 23 chromosomes**

They rapidly enter **Meiosis II**.

4 Spermatids

2 secondary spermatocytes

↓ Meiosis II

4 spermatids

So:

🧬 1 primary spermatocyte → 4 haploid spermatids

5 Spermiogenesis 🧑

Spermiogenesis = transformation of spermatids into mature spermatozoa **without further cell division**.

Important changes:

👉 **Acrosome forms** → contains enzymes important for penetrating the oocyte.

👉 **Nucleus condenses** → forms the sperm head.

⚡ **Flagellum/tail develops** → allows movement.

📄 **Mitochondria concentrate in the middle piece** → provide ATP.

✂️ Excess cytoplasm is removed.

➡️ **Spermatid → Spermatozoon**

⌚ Human spermatogenesis takes approximately **64–74 days**.

★ Easy sequence

Spermatogonium

↓ mitosis

Primary spermatocyte

↓ Meiosis I

Secondary spermatocyte ×2

↓ Meiosis II

Spermatids ×4

↓ spermiogenesis

👤 **Spermatozoa ×4**

🧑 O O G E N E S I S

Oogenesis = formation and maturation of the female gamete.

♀ **Site:** Ovary

🕒 **Begins:** Before birth

⚠️ Unlike spermatogenesis, oogenesis has long periods of meiotic arrest.

🧠 1 Prenatal period

Primordial germ cells differentiate into:

➡ Oogonia

Oogonia undergo **mitosis**, increasing their number.

By approximately the **5th month of fetal life**, most oogonia have either:

- ✗ Degenerated, or
- Differentiated into **primary oocytes**

The primary oocytes enter **Meiosis I**.

However...

- ⊖ They become arrested in **prophase I**, specifically the **diplotene stage**.

Primary oocyte = Prophase I arrest

🌸 2 From puberty onward

At puberty, during each ovarian cycle, a group of primary oocytes begins to resume meiosis.

Usually, **one dominant oocyte** completes Meiosis I.

Primary oocyte

⬇ Meiosis I

Produces:

- **Secondary oocyte**
- ◇ **First polar body**

- ⚠ The division is **unequal** because almost all the cytoplasm goes to the secondary oocyte.

○ 3 Secondary oocyte

The secondary oocyte immediately begins **Meiosis II**.

But...

- ⊖ It becomes arrested at:
Metaphase II

This occurs shortly before ovulation.
Therefore, the cell released during ovulation is:
☐ **Secondary oocyte arrested in metaphase II**

♥ 4 Fertilization

The secondary oocyte completes **Meiosis II ONLY if fertilization occurs.**

If sperm fertilizes the secondary oocyte:

Secondary oocyte

- ↓ completes Meiosis II
- ☐ **Mature ovum** + ☐ **Second polar body**

Then the male and female pronuclei fuse:

🧬 **23 + 23 = 46 chromosomes**

→ **Zygote**

🧬 MEIOSIS — WHY IS IT IMPORTANT?

[24/09/2026 6:55 AM] Imaanaa Harmee: Meiosis has **two major functions**:

1 Reduces chromosome number

46 → 23

This prevents chromosome number from doubling every generation.

💡 **Meiosis → 23**

💡 **Fertilization → 46**

2 Creates genetic variation 🧬🧬

Two major mechanisms:

☞ *A. Crossing over*

Occurs during **Prophase I**.

Homologous chromosomes exchange segments of DNA.

→ Creates **new combinations of genes**.

☞ *B. Independent assortment*

During meiosis I, maternal and paternal homologous chromosomes are distributed **independently and randomly**.

➡ Produces genetically different gametes.

⚠ NONDISJUNCTION

Nondisjunction = failure of chromosomes to separate properly during meiosis.

👤 Result:

Gametes may contain:

24 chromosomes → $n + 1$

22 chromosomes → $n - 1$

After fertilization, this can produce:

🧬 **Trisomy** → 47 chromosomes

🧬 **Monosomy** → 45 chromosomes

Example: Down syndrome

Trisomy 21

➡ Three copies of chromosome 21 instead of two.

[24/09/2026 6:55 AM] Imaanaa Harmee:

💧 **Spermatogenesis begins at puberty.**

💧 **Oogenesis begins before birth.**

💧 **Primary oocyte → Prophase I arrest.**

💧 **Secondary oocyte → Metaphase II arrest.**

💧 **Meiosis II in the secondary oocyte is completed only after fertilization.**

💧 **1 primary spermatocyte → 4 spermatozoa.**

💧 **1 primary oocyte → 1 functional ovum.**

💧 **Spermiogenesis = spermatid → spermatozoon, without cell division.**

👤 **Male:**

Puberty → 4 sperm → continuous

👤 **Female:**

Birth → 1 ovum → arrests twice → fertilization finishes

[24/09/2026 6:55 AM] Imaanaa Harmee:

🧬 **WEEK 1 OF HUMAN DEVELOPMENT**

 Days 1–7: From **Zygote** → **Blastocyst** → **Implantation**

The **first week** begins with **fertilization** and ends with the **blastocyst beginning implantation** into the uterine endometrium.

 **DAY 1 — FERTILIZATION**  

 **Usual site: Ampulla of the uterine tube**

Fertilization is the process by which the **sperm + secondary oocyte** unite to form a **zygote**.

 **Step 1: Capacitation**

Before a sperm can fertilize the oocyte, it undergoes **capacitation** in the female reproductive tract.

 Takes approximately **7 hours**.

Capacitation makes the sperm capable of:

-  undergoing the **acrosome reaction**
-  penetrating the oocyte's coverings

 **Step 2: Acrosome reaction**

The sperm's **acrosome** releases enzymes such as **acrosin** that help it penetrate the oocyte's protective layers.

 Sperm penetration occurs in sequence:

1  **Corona radiata** 



2  **Zona pellucida** 



3  **Oocyte plasma membrane**



 **Fusion**

 **BLOCK TO POLYSPERMY**

Once the first sperm fuses with the oocyte:

 Cortical granules are released



 Zona pellucida changes



⊘ Other sperm cannot penetrate

This prevents **polyspermy**.

★ Why is this important?

Normally:

1 oocyte + 1 sperm → 1 zygote

Not:

✗ 1 oocyte + multiple sperm

COMPLETION OF MEIOSIS II

Remember from oogenesis:

○ The secondary oocyte is arrested at **Metaphase II**.

After sperm entry:

➡ The secondary oocyte **completes Meiosis II**.

This produces:

○ Mature ovum

◇ Second polar body

 +  =  ZYGOTE

The genetic material forms:

 **Male pronucleus** = 23 chromosomes

+

 **Female pronucleus** = 23 chromosomes



 **Zygote = 46 chromosomes**

The zygote is therefore:

Diploid (2n = 46)

🌀 It contains the unique genetic combination of the new individual.

MITOCHONDRIAL DNA

An important genetic concept:

📄 **Mitochondrial DNA is inherited almost exclusively from the mother.**

Why?

Sperm mitochondria enter the oocyte only in limited amounts and are subsequently **eliminated/degraded**.

Therefore:

👤 Mother → mitochondrial DNA

👤 Father → essentially no mitochondrial DNA contribution

mtDNA inheritance = maternal inheritance

DAYS 2-3 — CLEAVAGE

After fertilization, the zygote begins rapid **mitotic divisions** called:

 **CLEAVAGE**

The individual cells produced are called:

👤 **BLASTOMERES**

Important:

The overall embryo **does not significantly increase in size** during cleavage because it remains enclosed within the **zona pellucida**.

Instead:

 Cell number ↑

 Cell size ↓

🕒 Approximate stages

Zygote

↓

👤 **2-cell** → ~30 hours

↓

👤 **4-cell** → ~40 hours

↓

👤 **12-16 cells** → Day 3

✳ **TOTIPOTENCY**

Early blastomeres have remarkable developmental potential.

🧬 Totipotent =

A cell capable of producing:

👤 **The entire embryo**

+

Extraembryonic tissues

Early blastomeres, particularly around the **2–8-cell stages**, have very high developmental potential.

COMPACTION

After several cleavage divisions, the blastomeres become tightly packed together.

This is called:

COMPACTION

Cells:

-  maximize contact with one another
-  develop tight junctions
-  become polarized
-  begin separating into inner and outer populations

A key adhesion molecule involved is **E-cadherin (uvomorulin)**.

[24/09/2026 6:55 AM] Imaanaa Harmee:  **DAY 3–4 — MORULA**

The embryo becomes a solid ball of approximately:

12–32 cells

This structure is called the:

MORULA

Morula = solid ball of cells 

It then enters the **uterine cavity**.

DAYS 4–5 — BLASTOCYST FORMATION

Fluid begins entering the morula.

 Fluid accumulation



Blastocyst cavity (blastocoel)



 **Blastocyst**

The blastocyst now has **two major cell populations**.

TWO PARTS OF THE BLASTOCYST

Embryoblast

📍 **Inner cell mass (ICM)**

➡ Forms the **embryo proper** 🧒

Trophoblast

📍 **Outer cell layer**

➡ Contributes to the **fetal part of the placenta** 🌱



Embryoblast → **Embryo** 🧒

Trophoblast → **placenta** 🌱

🕒 DAY 5 — HATCHING

Initially, the blastocyst is surrounded by the:

🕒 Zona pellucida

Around Day 5:

➡ Zona pellucida degenerates

➡ Blastocyst "hatches"

This is called:

🧒 HATCHING

Why is hatching necessary?

Because the blastocyst must be free from the zona pellucida to:

➡ expand

➡ attach to the endometrium

➡ begin implantation

🏠 DAYS 6–7 — IMPLANTATION

📅 Begins approximately **Day 6**

The blastocyst attaches to the:

Endometrium

Implantation usually occurs in the:

📍 **Posterior or anterior superior wall of the uterine body**

The **embryonic pole** faces the endometrium.

TROPHOBLAST DIFFERENTIATION

After implantation begins, the trophoblast differentiates into **two layers**:

1. *Cytotrophoblast*

- ✓ Inner layer
- ✓ Mononuclear
- ✓ Distinct cell boundaries
- ✓ Cells can divide

2. *Syncytiotrophoblast*

- ✓ Outer layer
- ✓ Multinucleated
- ✓ No distinct cell boundaries
- ✓ Highly invasive
- ✓ Invades the endometrium



CYTO = cells 😊

SYNCYTIO = fused/invasive layer 🗡️

SYNCYTIOTROPHOBLAST

The syncytiotrophoblast:

- ➡ Invades the endometrium
- ➡ Helps establish implantation
- ➡ Eventually contributes to the placenta
- ➡ Produces **hCG** 💉

★ hCG

Human chorionic gonadotropin (hCG) maintains the **corpus luteum** early in pregnancy.

The corpus luteum continues producing:

Progesterone

- ➡ Maintains the endometrium.

⚠ CLINICAL CORRELATIONS

1. "All-or-none" period

The first week is often described as a relatively **resistant/pre-implantation period**.

Exposure to severe teratogens may result in:

💀 **Death of the conceptus**

OR

☑ **Survival without a major structural malformation**

Therefore:

"All-or-none" phenomenon

⚠ However, this is a useful general rule rather than an absolute one; the exact effect depends on the **agent, dose, timing, and developmental stage**.

🚨 2. ECTOPIC PREGNANCY

Ectopic pregnancy = implantation of the conceptus **outside the normal uterine cavity**.

📍 Most commonly occurs in the:
Uterine tube

The **ampulla** is the most common tubal site.

Why is it dangerous?

As the pregnancy grows:

📏 Expansion of tube



💥 Tubal rupture



Severe internal hemorrhage



🚨 Potential maternal shock



Most common site of ectopic pregnancy = ampulla of uterine tube

[24/09/2026 6:55 AM] Imaanaa Harmee:

🧬 **WEEK 2 OF HUMAN DEVELOPMENT**

📅 Days 8–14 — **“WEEK OF TWOS”** 🤌

The **second week** is famous as the:

🤌 *WEEK OF TWOS*

Because many structures divide into **two layers** or **two cavities**.

★ The major “TWOS”:

 Trophoblast

✓ Cytotrophoblast + Syncytiotrophoblast

☺ **Embryoblast**

✓ Epiblast + Hypoblast

 Extraembryonic mesoderm

✓ Somatic + Splanchnic

Cavities

✓ Amniotic cavity + Yolk sac

○ **Embryonic disc**

✓ Bilaminar disc

 Chorionic villi

✓ Primary villi begin

Week 2 = TWO layers + TWO cavities + TWO mesoderm layers ✌

 DAY 8 — THE BILAMINAR DISK

The blastocyst is now **partially embedded** in the endometrium.

 1. Trophoblast → TWO layers

The trophoblast differentiates into:

⊗ **Cytotrophoblast**

✓ Inner layer

✓ **Mononuclear**

✓ Distinct cell boundaries

✓ Mitotically active

● **Syncytiotrophoblast**

✓ Outer layer

✓ **Multinucleated**

✓ No distinct cell boundaries

✓ Invasive

✓ Produces **hCG** 



CYTO = individual cells 🧑

SYNCYTIO = fused cells 🔗

🧑 2. Embryoblast → TWO layers

The embryoblast becomes the:

○ **BILAMINAR EMBRYONIC DISK**

🌀 Epiblast

- ✓ **Columnar cells**
- ✓ Located dorsally
- ✓ Faces the amniotic cavity
- ✓ Later gives rise to **all three germ layers** during gastrulation.

Hypoblast

- ✓ **Cuboidal cells**
- ✓ Located ventrally
- ✓ Faces the yolk sac
- ✓ Contributes to the lining of the yolk sac.

EPI = top ⬆️

HYPO = bottom ⬇️

3. AMNIOTIC CAVITY

A small cavity develops within the epiblast.



Expands into the:

AMNIOTIC CAVITY

It will eventually surround the developing embryo and contain **amniotic fluid**.

🏠 DAY 9 — LACUNAR STAGE

The blastocyst becomes **more deeply embedded** in the endometrium.

The implantation defect is temporarily closed by a:

Fibrin coagulum

Lacunae appear

Vacuoles develop within the:

🌀 **Syncytiotrophoblast**

These vacuoles fuse together:

➡ Large spaces called **lacunae**

This is called the:

LACUNAR STAGE

Syncytiotrophoblast → Vacuoles → Lacunae

◯ PRIMITIVE YOLK SAC

Hypoblast cells migrate and line the inside of the cytotrophoblast.

They form:

Exocoelomic (Heuser's) membrane

This encloses the:

EXOCOELOMIC CAVITY

also called the:

◯ PRIMITIVE YOLK SAC

DAYS 11–12 — UTEROPLACENTAL CIRCULATION

By Days 11–12:

☑ Blastocyst becomes **completely embedded** in the endometrium.

1. Maternal blood enters the lacunae

Syncytiotrophoblast lacunae connect with maternal:

Endometrial sinusoids

Maternal blood flows into the lacunar spaces.

➡ This establishes the beginning of:

PRIMORDIAL UTEROPLACENTAL CIRCULATION

★ This is an important early step toward placental circulation.

 *2. EXTRAEMBRYONIC MESODERM*

A new layer appears between:

● Trophoblast

and

Amnion/yolk sac

This is:

 EXTRAEMBRYONIC MESODERM

Large spaces develop within it.

These spaces fuse together:



EXTRAEMBRYONIC COELOM

also called:

CHORIONIC CAVITY



EXTRAEMBRYONIC MESODERM DIVIDES

The extraembryonic mesoderm is separated into:



Somatic extraembryonic mesoderm



Associated with trophoblast + amnion



Splanchnic extraembryonic mesoderm



Associated with yolk sac



Somatic = body wall side



Splanchnic = visceral side

[24/09/2026 6:55 AM] Imaanaa Harmee: **DAY 13 — CHORIONIC VILLI**

The implantation site is usually sealed by this time.

However, some women may experience slight:

Implantation bleeding

This can sometimes be mistaken for menstruation.



PRIMARY CHORIONIC VILLI

The cytotrophoblast forms cellular projections that extend into the syncytiotrophoblast.

These are:



PRIMARY CHORIONIC VILLI



Primary villi = cytotrophoblast core covered by syncytiotrophoblast.

Later, these villi become more complex and participate in placental development.



DEFINITIVE YOLK SAC

The primitive yolk sac becomes reorganized.



Smaller definitive yolk sac

The portions pinched off from the primitive yolk sac form:

Exocoelomic cysts

These may be found within the chorionic cavity.



DAY 14 — PROCHORDAL PLATE

The hypoblast develops a localized thickening called the:

★ PROCHORDAL PLATE

It marks the:

☹ **Future site of the mouth**

and helps establish:

➡ **Cranial (head) end**

➡ **Caudal (tail) end**

➡ **Cranial-caudal organization**

It is an important landmark for establishing the body's organization before gastrulation.

🦋 CONNECTING STALK

The embryo is suspended within the:

Chorionic cavity

by a structure called the:

🦋 CONNECTING STALK

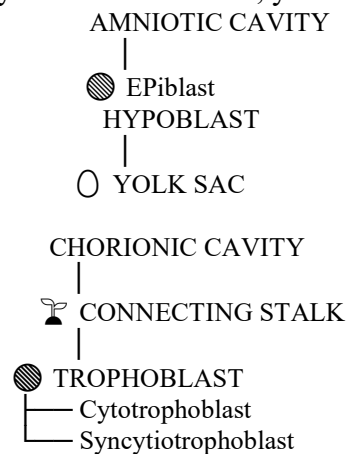
Later:

➡ Connecting stalk + developing vessels

➡ **Umbilical cord** 🦋

🧬 WHAT DOES THE WEEK 2 EMBRYO LOOK LIKE?

By the end of Week 2, you have:



And surrounding everything:

Maternal endometrium

🦋 hCG

The:

● Syncytiotrophoblast

produces:

hCG = Human Chorionic Gonadotropin

hCG:

- ➡ Maintains the **corpus luteum**
- ➡ Corpus luteum produces **progesterone**
- ➡ Progesterone maintains the **endometrium**
- ➡ Helps support early pregnancy 🧑

🔍 Pregnancy test

hCG can be detected in:

Blood → earlier

🚽 **Urine** → shortly afterward

★ This is why pregnancy tests detect **hCG**.

🚑 **CLINICAL: ECTOPIC PREGNANCY**

Ectopic pregnancy = implantation outside the normal uterine cavity.

📍 Most occur in the **uterine tube**, especially the **ampulla**.

🚑 *Why dangerous?*

Growing embryo



Tubal expansion



💥 Possible rupture



Internal hemorrhage



⚠️ Maternal emergency

🍽️ **NUTRITION DURING WEEK 2**

Early in Week 2, the embryo is very small and has no mature placental circulation.

Nutrition is obtained primarily from:

Maternal endometrial tissues



through **diffusion**

As the trophoblast and placental circulation develop, maternal–embryonic exchange becomes increasingly organized.

👉 WEEK OF TWOS

Trophoblast

➡ Cytotrophoblast + Syncytiotrophoblast

Embryoblast

➡ Epiblast + Hypoblast

Extraembryonic mesoderm

➡ Somatic + Splanchnic

Two major cavities

➡ Amniotic cavity + Yolk sac

★ Remember these:

🧬 Epiblast → all 3 germ layers

🌀 Syncytiotrophoblast → invasion + hCG

Lacunae → maternal blood

🌱 Primary chorionic villi → begin placental development

🌀 Prochordal plate → future mouth/head end

🌱 Connecting stalk → future umbilical cord

Extraembryonic coelom = chorionic cavity

○ Hypoblast → yolk sac lining

“TWO layers, TWO cavities, TWO mesoderm layers, and trophoblast starts building the placenta.” 👉 🧬 🌱

🧬 Week 1 → **Implantation begins** 🏠

👉 Week 2 → **Bilaminar disc + placenta starts** 🌱

✳️ Week 3 → **Gastrulation + trilaminar disc** 💧

[24/09/2026 6:55 AM] Imaanaa Harmee:

🧬 **WEEK 3 OF HUMAN DEVELOPMENT**

✳️ **GASTRULATION**

Gastrulation is the process that transforms the:

Bilaminar embryonic disc 👉

➡ **Trilaminar embryonic disc** ③

This creates the **three primary germ layers**:

● Ectoderm



Nervous system 🧠

Epidermis 🏠

● Mesoderm



Muscle 🦠

Bone 🦴

Connective tissue

Blood ❤️

Endoderm



GI tract lining 🍷

Respiratory tract lining

Many glands

PRIMITIVE STREAK

Gastrulation begins with formation of the:

★ PRIMITIVE STREAK

It appears on the **dorsal surface of the epiblast**, initially near the **caudal end**.

It contains:

— Primitive groove

★ Primitive node

● Primitive pit



Primitive node = organizer 🎯

It helps establish the **body plan and major axes** of the embryo.

INVAGINATION

Epiblast cells migrate toward the primitive streak.

Then they:



Move inward



Detach



Migrate beneath the epiblast

This process is called:

INVAGINATION

🧬 **FGF8** is an important signaling molecule involved in regulating epiblast cell migration during gastrulation.

🧬 FORMATION OF THE THREE GERM LAYERS

📌 First wave

Epiblast cells migrate inward and **replace the hypoblast**.

➡ **Embryonic endoderm**

📌 Second wave

Other epiblast cells migrate between the new endoderm and remaining epiblast.

➡ **Mesoderm** 🌀

📌 Remaining epiblast cells

➡ **Ectoderm** 🌀

★ **CRITICAL POINT:**

ALL THREE germ layers originate from the EPIBLAST.

🌀 NOTOCHORD FORMATION

Another major event of Week 3 is formation of the:

🧬 NOTOCHORD

Cells from the primitive node migrate **cranially** and form the **notochordal process**, which ultimately becomes the definitive notochord.

What does the notochord do? 🌀

It:

- ✓ Establishes the **longitudinal axis** of the embryo
- ✓ Provides structural organization
- ✓ Acts as an important signaling center
- ✓ Induces development of the **neural plate** 🧠
- ✓ Contributes to the future axial skeleton

★ **Clinical/remnant:**

Most of the notochord disappears, but remnants persist as the:

Nucleus pulposus of intervertebral discs 🦴

🌀 ESTABLISHMENT OF BODY AXES

Week 3 establishes the basic body plan.

🌀 Major axes:

- ◇ **Cranial ↔ Caudal**
- ◇ **Right ↔ Left**
- ◇ **Dorsal ↔ Ventral**

The embryo also establishes:

↔ **Bilateral symmetry**

Important structures include:

- 🌀 Primitive streak
- 🌀 Primitive node
- 🌀 Notochord
- 🌀 Prochordal plate

👄 OROPHARYNGEAL & CLOACAL MEMBRANES

At specific locations, **ectoderm and endoderm remain directly attached** with no intervening mesoderm.

👄 Cranially:

Oropharyngeal membrane

➡ Future **mouth**

👄 Caudally:

Cloacal membrane

➡ Future **anal/urogenital opening region**

😊:

ORO = mouth 👄

CLOACA = posterior opening

[24/09/2026 6:55 AM] Imaanaa Harmee: 🧠 NEURULATION

Near the end of Week 3, development of the CNS begins.

🧬 Notochord

⬇ induces

🌀 Neural plate

⬇

🧠 Neural groove + neural folds

⬇

🧠 Neural tube

This process is called:

★ NEURULATION

🧠 Neural plate

The ectoderm becomes thickened over the notochord.

➡ Neural plate

The edges elevate:

⬆ Neural folds

The depression between them:

⬇ Neural groove

The folds then approach each other and fuse:

🌀 Neural tube

★ Neural tube → CNS

🧠 Brain + spinal cord

🧬 NEURAL CREST CELLS

As the neural folds fuse, cells at their edges become:

★ Neural crest cells

These migrate throughout the embryo and contribute to many structures, including:

💓 Autonomic ganglia

🧠 Sensory ganglia

👤 Melanocytes

🦷 Some craniofacial structures

⚡ Adrenal medulla

💧 Neural crest = **highly migratory cells with many derivatives.**

🔑 SOMITE FORMATION

Around **Day 20**, the **paraxial mesoderm** begins dividing into paired blocks:

🧱 SOMITES

They appear on both sides of the neural tube.

🕒 Rate:

Approximately **3 pairs/day** initially.

SOMITE DERIVATIVES

Each somite differentiates into:

 Sclerotome

 **Vertebrae + ribs**

 Myotome

 **Skeletal muscle**

 Dermatome

 **Dermis of skin**

MISSED PERIOD

The beginning of Week 3 approximately corresponds to the time when menstruation would normally be expected.

Therefore:

No menstruation



 Missed period



 Pregnancy test

can become the first indication that pregnancy has occurred.

[24/09/2026 6:55 AM] Imaanaa Harmee:

EMBRYONIC PERIOD

Weeks 3–8 — THE PERIOD OF ORGANOGENESIS

The **embryonic period** extends from the **beginning of Week 3 to the end of Week 8**.

It is the period when:

 Germ layers differentiate

 Neural tube develops

 Cardiovascular system develops

 Skeleton begins forming

 Muscles develop

 Limbs and face develop

Major organ systems are established

By the end of Week 8, the embryo has a **distinctly human appearance** and enters the **fetal period**.

WHY ARE WEEKS 3–8 SO IMPORTANT?

☢ MAXIMUM TERATOGEN SUSCEPTIBILITY

During Weeks **3–8**, major organs are actively forming.
Therefore, exposure to teratogens such as:

- 💊 Certain drugs
- 🍷 Alcohol
- 🦠 Infections
- ☢ Radiation
- 🧪 Chemicals

can cause:

- 🏰 Major structural congenital abnormalities

🧠 Remember:

Week 1 → "All-or-none" ● ○

Weeks 3–8 → **Major structural defects** 🏰

Fetal period → **Growth + functional maturation** 📊

🌸 WEEK 3 — GASTRULATION

The most characteristic event of Week 3 is:

🧬 GASTRULATION

Bilaminar embryonic disc



Trilaminar embryonic disc

Three germ layers form:

- 🌀 **Ectoderm**
- 🌀 **Mesoderm**
- Endoderm**

All three germ layers originate from the EPIBLAST.

— PRIMITIVE STREAK

Gastrulation begins with formation of the:

- ★ Primitive streak

It appears on the dorsal surface of the epiblast.

Epiblast cells migrate through the primitive streak:

- ➡ Endoderm
- ➡ Mesoderm
- ➡ Remaining epiblast → Ectoderm

NOTOCHORD

The notochord develops during Week 3.

Its major functions include:

-  Establishes the body's longitudinal axis
-  Induces development of the neural plate
-  Provides the developmental basis for the axial skeleton

★ Adult remnant:

Notochord → **nucleus pulposus**

WEEK 3 — CARDIOVASCULAR DEVELOPMENT

The cardiovascular system begins developing very early.

Vasculogenesis

Vasculogenesis = formation of blood vessels from precursor cells.

Early blood islands develop in the:

 **Yolk sac**

and later within the embryo.

 The primitive heart tube develops and begins functioning very early.

WEEK 4 — NEURULATION

The neural plate forms toward the end of Week 3.

Then:

-  Neural plate
- 
-  Neural folds
- 
-  Neural groove
- 
-  Neural tube

This process is:

 **NEURULATION**

★ Neural tube → CNS

NEURAL TUBE CLOSURE

The neural tube closes through fusion of the neural folds.

 Important approximate dates:

🧠 **Anterior neuropore → Day 25**

🧠 **Posterior neuropore → Day 27**

Failure of closure can cause:

🔔 **Neural tube defects**

Examples:

🧠 **Anencephaly**

🔑 **Spina bifida**

🏗 SOMITES

Beginning around:

📅 **Day 20**

The **paraxial mesoderm** segments into:

🏗 **SOMITES**

They develop on both sides of the neural tube.

Initially, approximately:

3 pairs/day

are formed.

Eventually, up to **42–44 pairs** may form, although only about **35 pairs persist** as recognizable somites.

[24/09/2026 6:55 AM] Imaanaa Harmee: 📄 WEEKS 4–8 — LIMB DEVELOPMENT

👉 **Upper limbs**

Appear first:

📅 **Day 26–27**

👉 **Lower limbs**

Appear slightly later:

📅 **Day 28–30**

So:

👉 **Upper limb first → Lower limb second**

👉 WEEK 6 — DIGITAL RAYS

During Week 6:

The hand and foot plates develop:

🔖 **Digital rays**

These are the early indications of:

👉 **Fingers**

 Toes

WEEK 8 — DIGIT SEPARATION

Initially, the developing digits are connected by tissue.

Then:

 **Programmed cell death (apoptosis)**

removes the tissue between them.



 Separate fingers

 Separate toes

FACE DEVELOPMENT

During Weeks 4–8, facial prominences develop and fuse.

They contribute to:

 Nose

 Upper lip

 Face

 Primary palate

 Upper lip

Develops through fusion of facial prominences, particularly the:

✓ Maxillary prominences

✓ Medial nasal prominences

EYE POSITION

Initially:

 Eyes are positioned more **laterally**.

As the face develops:

 Eyes move toward a more **anterior/ventral position**.

This contributes to the characteristic human facial appearance.

WEEK 6 — PHYSIOLOGICAL HERNIA

The midgut grows extremely rapidly.

 Midgut growth



 Abdominal cavity temporarily too small



Midgut herniates into the **umbilical cord**

This is called:

★ PHYSIOLOGICAL MIDGUT HERNIATION

⚠ Normal!

It occurs because the intestine grows faster than the abdominal cavity.

Later, the intestines return to the abdominal cavity.

🔑 SKELETAL & MUSCULAR DEVELOPMENT

Somites contribute significantly to the developing musculoskeletal system.

🔑 Sclerotome

➡ Vertebrae + ribs

🔗 Myotome

➡ Skeletal muscles

🏠 Dermatome

➡ Dermis

By the end of Week 8, the developing axial skeleton and musculature have undergone substantial differentiation.

👤 END OF WEEK 8

By the end of the embryonic period:

🧠 CNS established

❤ Cardiovascular system developed

🔑 Skeleton developing

🔗 Muscles developing

👉 Limbs formed

👁 Eyes developing

👄 Face developing

Gut has undergone physiological herniation and begins returning

And most importantly:

👤 THE EMBRYO HAS A DISTINCT HUMAN APPEARANCE

👉 Embryonic period ends



📅 FETAL PERIOD BEGINS

Week 9 → Birth

Main focus:

Growth + maturation + functional development

HOW IS EMBRYONIC AGE MEASURED?

During early development, age can be described using:

 **Somite number**

Later, particularly during Weeks 5–8:

 **Crown-rump length (CRL)**

is commonly used.

★ CRL

Measurement from:

Top of head → bottom of trunk

It is an important method for estimating embryonic/fetal age.

[24/09/2026 6:55 AM] Imaanaa Harmee:  **FETAL PERIOD**

 Week 9 → Birth

The **fetal period** begins at the **start of Week 9** and continues until birth.

 **Embryonic period = Build the organs** 

 **Fetal period = Grow + mature the organs** 

During this period, the organs that formed during Weeks 3–8 continue to:

 Increase in size

 Differentiate

 Mature functionally

 Gain weight

MAJOR PHYSICAL CHANGES

HEAD GROWTH SLOWS

At the beginning of the fetal period, the head is **disproportionately large**.

 Relative head size:

3rd month:  $\approx \frac{1}{2}$ of CRL



5th month:  $\approx \frac{1}{3}$ of CHL



Birth:  $\approx \frac{1}{4}$ of CHL

 The fetus gradually develops more **body proportions resembling a newborn**.

FACIAL DEVELOPMENT

During the **3rd month**:

👁️ Eyes move from a more lateral position → toward the front/ventral aspect.

👂 Ears move toward their definitive position on the sides of the head.

👃 Nose becomes more prominent.

➡️ The face becomes increasingly **human-looking**. 😊

👶 SKIN DEVELOPMENT

Around the 6th month:

The skin is:

🟡 Reddish

〰️ Wrinkled

👶 Thin

because there is relatively little **subcutaneous tissue/fat**.

Later:

⬇️ Subcutaneous fat increases

⬇️ Skin becomes smoother

📖 VERNIX CASEOSA

Toward the end of pregnancy, fetal skin becomes covered by:

👶 **VERNIX CASEOSA**

A

○ Whitish

📖 Fatty

🛡️ Protective substance.

Functions:

🛡️ Protects fetal skin from prolonged exposure to amniotic fluid.

👶 BODY FAT & WEIGHT

📏 Length:

Most rapid increase occurs during:

Months **3+4+5**

⚖️ Weight:

Major weight gain occurs particularly during:

Last 2 months 📊

because of increasing:

 Subcutaneous fat

 Gives the fetus a more **rounded appearance**.

MONTH BY MONTH FETAL DEVELOPMENT

3rd MONTH

 Weeks 9–12

Several important changes occur.

 Ossification

Primary ossification centers appear in:

 Long bones

 Skull

by approximately **Week 12**.

 External genitalia

External genitalia become sufficiently developed that **fetal sex can often be determined by ultrasound**.

Intestines return

Week 6:

Midgut → physiological herniation

During the **10th week**, the intestinal loops return to the abdominal cavity.

 **return occurs around Week 10, not simply anywhere in the third month.**

4th–5th MONTH

 Weeks 13–20

 Rapid length growth

The fetus becomes much longer.

LANUGO

Fine, soft fetal hair appears over the body:

✦ **LANUGO**

Also:

👁 Eyebrows develop

 Scalp hair becomes visible

QUICKENING

Quickening = first fetal movements felt by the mother

Usually occurs around:

 **18–20 weeks**

 Commonly during the **5th month**.

6th MONTH

 Weeks 21–24

The fetus continues rapid growth and organ maturation.

However:

Respiratory system

 CNS

are still immature.

Therefore:

 A fetus born very early during this period has a **low chance of survival**, although modern neonatal care can improve outcomes, especially closer to 24 weeks.

7th MONTH

 Weeks 25–28

Major maturation continues.

The fetus may reach approximately:

 **~1,100 g**

and:

 **~25 cm CRL**

Important:

Lung development and CNS maturation have progressed significantly.

 **Survival after premature birth becomes much more likely**, although substantial medical support is usually required.

8th–9th MONTH

 Weeks 29–38

Now the fetus mainly undergoes:

 Rapid weight gain

 Fat deposition

Lung maturation

 CNS maturation

 Muscle development

Approximately **half of the full-term weight** may be gained during the final weeks of pregnancy.

 SKULL DEVELOPMENT

The skull reaches a large circumference during late pregnancy.

Why?

Because the:

 Brain grows rapidly.

The skull must accommodate this growth.

 AT BIRTH

A typical full-term newborn weighs approximately:

 **3,000–3,400 g**

Male newborns:

♂ **Testes should normally have descended into the scrotum.**

[24/09/2026 6:55 AM] Imaanaa Harmee:

 PLACENTA

The placenta has **two major components**:

 *Fetal portion*

Derived mainly from:

✓ **Chorion frondosum** 

 *Maternal portion*

Derived from:

Decidua basalis



Fetal = Chorion frondosum

Maternal = Decidua basalis

AMNION + CHORION

As pregnancy progresses:

Amnion grows



 Chorion grows



 They fuse

This eventually **obliterates the uterine cavity** as a distinct space.

AMNIOTIC FLUID

Amniotic fluid surrounds the fetus.

★ Functions:

 **Protection** — cushions the fetus

 **Movement** — allows free fetal movement

 **Prevents adhesion** — prevents the fetus from sticking to surrounding tissues

 **Helps maintain a stable environment**

Supports lung development

 *FETUS SWALLOWS AMNIOTIC FLUID*

The fetus:

Amniotic fluid



 Swallows it



 GI tract



Absorption



Fetal kidneys



 Urine



Amniotic fluid

So fetal urine contributes significantly to the amniotic fluid later in pregnancy.

Fetal urine is a major source of amniotic fluid in later pregnancy.

The placenta also participates in fluid exchange.

⚠ TERATOGENS DURING THE FETAL PERIOD

Compared with Weeks 3–8:

 Teratogen susceptibility decreases

Because the major organ structures have already been established.

Therefore, exposure is more likely to cause:

⚙️ **Functional abnormalities**

📏 **Growth restriction**

🧠 **Neurological effects**

rather than major structural malformations.

⚠️ But this does **NOT** mean the fetus is resistant to teratogens.

Some structures continue developing throughout pregnancy, especially the:

🧠 CNS

Lungs

👁️ Eyes

so harmful exposures can still cause serious abnormalities.

📅 DATING THE PREGNANCY

There are two important ways to think about pregnancy duration.

👤 **From LNMP**

Last Normal Menstrual Period

➡️ Approximately:

280 days = 40 weeks

🧬 **From fertilization**

Actual developmental age:

~266 days = 38 weeks

🧠 Why the difference?

Pregnancy is conventionally dated from the **first day of the last menstrual period**, which is about **2 weeks before ovulation and fertilization** in a typical cycle.

LNMP → 40 weeks

Fertilization → 38 weeks

[24/09/2026 6:56 AM] Imaanaa Harmee:

👶👶 **MULTIPLE PREGNANCY & TWINS**

Multiple pregnancy = development of **more than one fetus** during a single pregnancy.

The most common type is:

👶👶 TWINS

Other possibilities include:

👶👶👶 Triplets

👶👶👶👶 Quadruplets

👶👶👶👶👶 Quintuplets

 Multiple pregnancies have become more common partly because of **assisted reproductive techniques and fertility treatments**.

TWO MAJOR TYPES OF TWINS

DIZYGOTIC TWINS

  "Fraternal twins"

Formed from:

 **2 secondary oocytes**

+

 **2 sperm**



 **2 separate zygotes**

Therefore:

Two fertilizations → two embryos

 Genetic relationship

Dizygotic twins are **genetically different**, just like ordinary siblings.

They can be:



because they result from different eggs and sperm.

MEMBRANES OF DIZYGOTIC TWINS

Each embryo develops its own:

 **Chorion**

Amnion

 **Placenta**

Therefore:

 **DIAMNIOTIC–DICHORIONIC**

2 amnions + 2 chorions

The two placentas may become so close that they appear **fused**, but developmentally they are still separate placentas.

MONOZYGOTIC TWINS

👶👶 "Identical twins"

Formed from:

○ 1 secondary oocyte

+

🧬 1 sperm

↓

🧬 1 zygote

↓

✂️ Embryo splits into two

Therefore:

One fertilization → one zygote → two embryos

Because they originate from the same zygote, they are usually **genetically very similar/essentially genetically identical**.

The **timing of separation** determines the fetal membranes.

🕒

✂️ **EARLY split → more separate membranes**

🕒 **LATE split → more shared membranes**

📅 MONOZYGOTIC TWIN SPLITTING

✂️ *EARLY SPLIT*

📅

Around the 2–3 cell stage

The zygote separates very early.

Each embryo develops:

🌱

Separate chorion

Separate amnion

🌱

Separate placenta

★ Result:

Dichorionic–diamniotic

2 chorions + 2 amnions

👉

Looks like dizygotic twins in terms of membranes.

✂️ *SPLIT AT INNER CELL MASS*



Approximately Days 4–8

The inner cell mass separates into two embryonic groups.

They share:



One chorion



One placenta

but each has:

Its own amniotic sac



Result:

Monochorionic–diamniotic

1 chorion + 2 amnions



This is the **most common type of monozygotic twinning**.



LATE SPLIT



Approximately Days 8–13

The separation occurs after formation of the bilaminar embryonic disc.

The twins share:



1 chorion

1 amnion



1 placenta



Result:

Monochorionic–monoamniotic

1 chorion + 1 amnion



These pregnancies have increased risks, including **umbilical cord entanglement**.



Earlier split = more independent

Later split = more shared structures

Think:

EARLY ✂



2 placenta

2 chorion

2 amnion



MID ✂



1 placenta

1 chorion
2 amnion


LATE ✂

↓
1 placenta
1 chorion
1 amnion


 *VERY LATE / INCOMPLETE SPLIT*

If the separation is **incomplete**, the embryos remain physically connected.

➡ **Conjoined twins** 

This occurs when the embryonic disc does not completely separate.

[24/09/2026 6:56 AM] Imaanaa Harmee:

 **PLACENTA**

 The Mother–Fetus Exchange Organ

The **placenta** is a temporary, discoid-shaped organ that connects the mother and fetus.

★ Dual origin:

 **Maternal part → Decidua basalis**

 **Fetal part → Chorion frondosum**

Maternal = Decidua basalis

Fetal = Chorion frondosum

The placenta performs:

Gas exchange

 Nutrient exchange

 Waste removal

 Hormone production

 Passive immunity

 1. MATERNAL PORTION

 **Decidua basalis**

The **decidua basalis** is the modified endometrium located between:

Blastocyst/placenta ↔ Myometrium

During implantation, the endometrial stromal cells undergo:

★ Decidual reaction

Cells enlarge and accumulate:

-  Lipids
-  Glycogen

These provide nutritional support during early pregnancy.

2. FETAL PORTION

Chorion frondosum

The fetal component develops from the **villous chorion**.

It contains:

-  Tertiary chorionic villi

These contain fetal:

- Blood vessels
-  Connective tissue
-  Fetal capillaries

The fetal surface is covered by:

-  **Chorionic plate**

and:

Amnion

Therefore, the fetal surface is:

-  Smooth
-  Shiny
-  Bluish

and contains large fetal blood vessels.

INTERVILLOUS SPACE

This is a space surrounding the chorionic villi.

It contains:

-  Maternal blood

Blood enters through:

Spiral arteries

and surrounds the fetal chorionic villi.



Maternal blood and fetal blood normally **do NOT directly mix**.

Substances cross the **placental membrane**.

PLACENTAL MEMBRANE

★ Placental barrier

The placental membrane separates:

 **Maternal blood**

from:

 **Fetal blood**

It becomes progressively **thinner** as pregnancy advances.

EARLY PREGNANCY

Initially, the placental membrane contains approximately **4 layers**:

- 1  **Syncytiotrophoblast**
- 2  **Cytotrophoblast**
- 3  **Connective tissue of villus**
- 4  **Fetal capillary endothelium**

LATE PREGNANCY

The barrier becomes much thinner.

★ To increase efficiency of exchange

The fetal capillaries move closer to the surface and much of the cytotrophoblast disappears.

The main remaining layers are:

Syncytiotrophoblast

 **Fetal capillary endothelium**

 **Shorter diffusion distance = more efficient exchange**

WHAT CROSSES THE PLACENTA?

O₂

CO₂

 **Glucose**

 **Amino acids**

 **Many drugs**

 **Water/electrolytes**

IgG antibodies

★ IgG

Maternal IgG crosses the placenta:

 Mother →  Fetus

 Provides:

 Passive immunity

WHAT GENERALLY DOES NOT CROSS EASILY?

Large molecules such as:

- ✗ Many proteins
- ✗ Most bacteria
- ✗ Maternal cells

For example:

 Insulin

Insulin is a large protein hormone and **does not normally cross the placenta in significant amounts**.

 However, don't interpret the placental barrier as an absolute wall—some pathogens, antibodies, drugs, and other molecules can cross.

FUNCTIONS OF THE PLACENTA

Exchange + Endocrine + Immunity

METABOLIC/EXCHANGE FUNCTION

Gases

Mother → Fetus:

O₂

Fetus → Mother:

CO₂

Nutrients

Placenta transfers:

-  Glucose
-  Amino acids
-  Fatty acids
-  Vitamins
-  Electrolytes
-  Water

Waste

Fetus → Mother:

- ✎ CO₂
- ✎ Urea
- ✎ Creatinine
- ✎ Other metabolic wastes

Mother's circulation then removes them.

ENDOCRINE FUNCTION

The placenta acts as a major endocrine organ.

★ Important placental hormones:

- ✎ hCG
- ✎ hPL
- ✎ Progesterone
- ✎ Estrogens

hCG

★ Human Chorionic Gonadotropin

Produced mainly by:

Syncytiotrophoblast

Its major early function:

- 👤 hCG
- ⬇️
- Maintains corpus luteum
- ⬇️
- ✎ Progesterone production continues
- ⬇️
- 🏠 Endometrium maintained

It is also the hormone detected by:

- ✎ Pregnancy tests

hPL

Human Placental Lactogen

Produced by:

Syncytiotrophoblast

It contributes to maternal metabolic adaptation and helps ensure nutrients are available for the fetus.

It promotes:

- ⬆️ Maternal lipolysis

- ⬆ Maternal insulin resistance
 - ➡ More glucose becomes available for the fetus.
- [24/09/2026 6:56 AM] Imaanaa Harmee: **PROGESTERONE**

Important functions:

- 🏠 Maintains endometrium
- 👶 Supports pregnancy
- 🔴 Reduces uterine contractions

As pregnancy progresses, the placenta becomes the major source of progesterone.

ESTROGENS

Estrogens contribute to:

- 📏 Growth of uterus
- Increased uterine blood flow
- 👶 Breast development

IMMUNOLOGICAL FUNCTION

Maternal:

IgG

crosses the placenta.

👶 IgG → 🧒

- ➡ Provides **passive immunity** to the fetus/newborn.
- This protection can continue for some time after birth.

FETAL MEMBRANES

The fetal membranes include:

- Amnion
- 🌿 Chorion
- Yolk sac
- 🌿 Allantois

AMNION

The amnion is a thin membrane surrounding the:
Amniotic cavity

which contains:

- 💧 Amniotic fluid

FUNCTIONS OF AMNIOTIC FLUID

- 🛡 Protection

Acts as a cushion against mechanical injury.

 Movement

Allows the fetus to move freely.

 Musculoskeletal development

Movement helps normal development of muscles and joints.

Prevents adhesion

Prevents the fetus from sticking to surrounding tissues.

Lung development

Fetal breathing movements and amniotic fluid are important for normal pulmonary development.

WHERE DOES AMNIOTIC FLUID COME FROM?

Early in pregnancy:

 Much of the fluid comes from **maternal sources**.

Later:

 Fetal urine becomes a major contributor.

The fetus also:

 Swallows fluid

and it is absorbed through the fetal GI tract.



Fetal urine is a major source of amniotic fluid later in pregnancy.

AMNIOCHORIONIC MEMBRANE

As the fetus grows:

Amnion expands



Contacts the smooth chorion



They fuse



The **chorionic cavity is obliterated**



Together they form the:

"BAG OF WATERS"

UMBILICAL CORD

The umbilical cord connects:

 Fetus ↔  Placenta

Normally it contains:

★ 3 VESSELS

2 umbilical arteries

1 umbilical vein

UMBILICAL ARTERIES

2 arteries

Carry:

 Fetus →  Placenta

Therefore they carry:

 Oxygen

 CO₂

 Metabolic waste

Umbilical arteries = deoxygenated blood

⚠ They are arteries because they travel **away from the fetal heart**, not because they carry oxygen-rich blood.

UMBILICAL VEIN

1 vein

Carries:

 Placenta →  Fetus

Therefore:

 Oxygen

 Nutrients

Umbilical vein = oxygenated blood

Again, it's a **vein** because it travels **toward the fetal heart**.

WHARTON'S JELLY

The umbilical vessels are embedded in:

★ **Wharton's jelly**

A gelatinous connective tissue.

Function:

- Protects the umbilical vessels from compression.